

Introduction to Software Engineering (F28SD) 2024-25

Coursework-1

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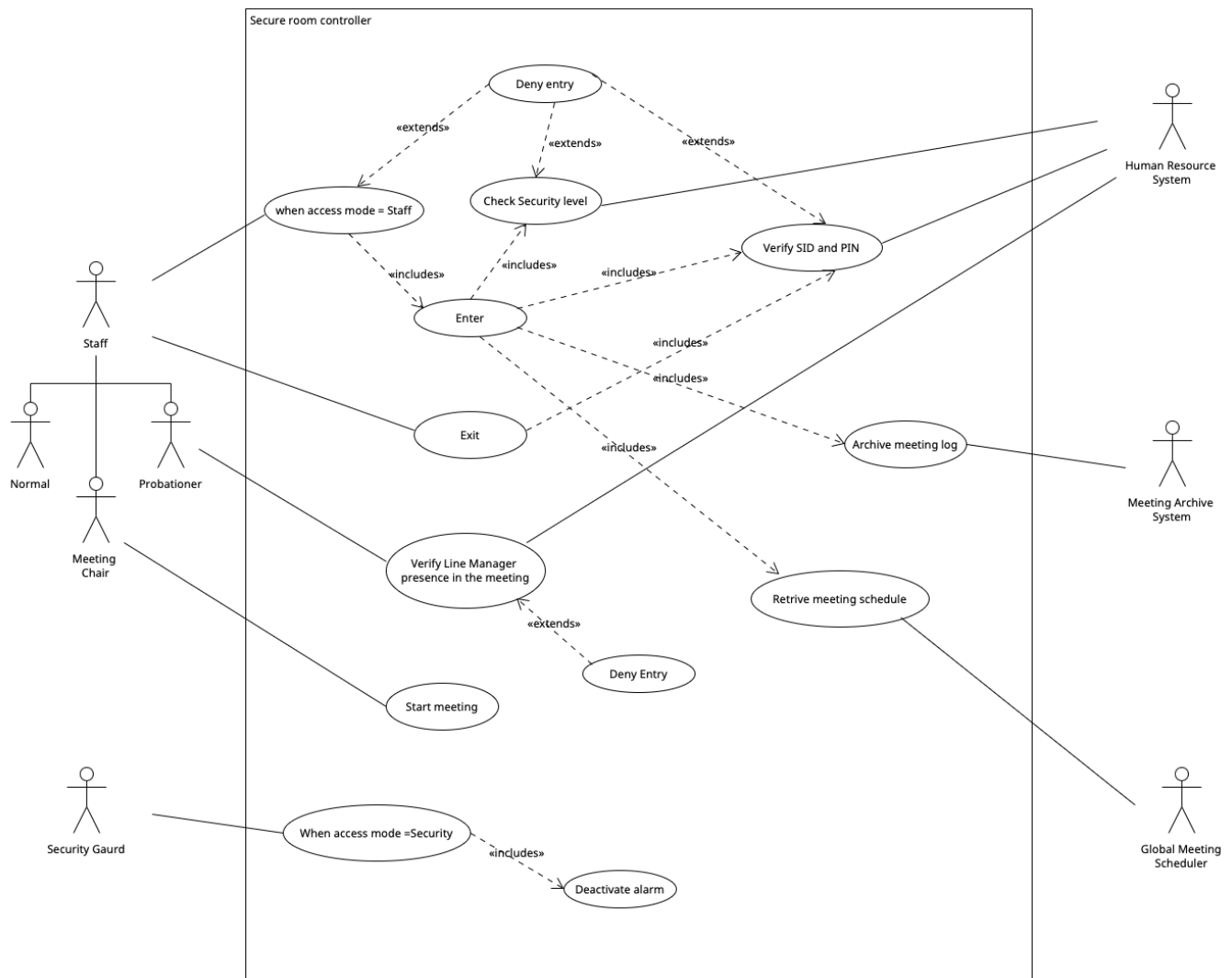
D1: Gaps and assumptions

Sno.	Gaps	Assumption
1.	No specification on what happens when a person enters an incorrect SID or PIN multiple times.	If a staff member enters an incorrect SID or PIN more than three times, their access is denied, and security is alerted.
2.	Unclear how security guards are notified about suspicious entry attempts	Security guards receive an automated alert through their monitoring system when unauthorized access attempts are detected.
3.	No mention of how SRC handles staff who have been terminated but still possess a valid SID and PIN	Staff are denied entry, and their status is updated in the HRS system.
4.	Unclear of what happens when Scheduled Meeting ends but Participants haven't left.	Participants are given a 10-minute grace period to exit. Failure to leave triggers a security notification.
5.	No defined response if a staff member tries to enter when no meeting is scheduled.	They should be denied entry.
6.	No clarification on what happens if a staff member with insufficient security clearance tries to attend a meeting.	They should be denied entry.
7.	what happens mc tries to leave due to emergency.	The MC role is reassigned to some other staff present in the meeting
8.	Unclear process for updating staff security clearance levels.	Any security level change must be requested by SRC and updated in the HRS system.
9.	No defined action if a probationary staff member's line manager leaves the meeting.	If the line manager leaves, the probationer is automatically notified and required to exit as well.
10.	No contingency plan for airlock door failure.	The door should be regularly maintained.
11.	No specification on whether probationers are assigned a Security Clearance Level (SCL).	Probationers are assigned an SCL with additional restrictions compared to regular staff.
12.	Unclear if verification is required to open the emergency exit.	The emergency exit operates manually and can be opened without authentication in case of an emergency.

D2: Functional and Non-functional requirements

ID	Description	Priority
FR1	The SRC shall recognise the two access modes	C
FR2	The SRC shall validate staff credentials using HRS before granting access.	M
FR3	The SRC shall deny access to a staff member if their security clearance level (SCL) is lower than required for the meeting.	M
FR4	The SRC shall prevent entry if a staff member is not listed in the meeting schedule.	M
FR5	The SRC shall allow only one person in the airlock at a time.	M
FR6	The SRC shall enforce that a probationer must be accompanied by their line manager during top secret and secret meetings.	S
FR7	The SRC shall verify that a meeting chair (MC) is present before marking a meeting as started.	S
FR8	The SRC shall trigger a silent alarm if unauthorized access is detected.	M
FR9	SRC shall retrieve a daily Meeting Schedule (MS) from the external Global Meeting Scheduler (GMS) at 02:00 Scheduler (GMS) at 02:00	M
FR10	The SRC shall generate a Meeting Log (ML) for each meeting.	M
FR11	The ML shall be archived to the external Meeting Archive System (MAS) at 01:59 daily. Once MAS acknowledges receipt, SRC shall delete its local ML copy.	M
FR12	The SRC shall allow meetings only between 06:00 and 22:00.	M

D3: Use Case diagram



D4: Use case specifications

Use Case: Enter staff
ID: 1
Goal: Allow authorized staff members to enter a secure room
Primary actor: Staff
Secondary actor(s): Human Resource system (HRS)
Preconditions: 1. The staff member must be registered in HRS with a valid SID and PIN. 2. The staff's Security clearance level must meet or exceed the Meeting security level .
Postconditions: 1. If access is granted, the system updates the meeting log (ML) with the staff's entry.
Main flow: 1. The staff selects "Enter Room" on the touchscreen. 2. The system prompts for SID and PIN. 3. The SRC sends SID & PIN to HRS for validation. 4. If valid, SRC retrieves the staff's Security clearance level from HRS. 5. SRC checks the meeting's Security level 6. If access is valid, checks if staff is a probationer. 7. If yes checks if its line manager is present. 8. If line manger is present, airlock door opens.
Alternative flows: 3a. Invalid SID or PIN 1. SRC denys entry if user has invalid SID and PIN 2. Use case is terminated. 5a. Insufficient security clearence 1. SRC denys entry if required security level is not met. 2. Use case is terminated. 6a. Probationer Handling 1. If the staff is a probationer and line manager is not present, SRC denys entry. 2. Use case is terminated.

Use Case: Exit staff
ID: 2
Goal: Allow staff to leave the secure room .
Primary actor: Staff
Secondary actor(s): None
Preconditions: 1. No silent alarm should be active.
Postconditions: 1. If exit is granted, the system updates the meeting log (ML) with the staff's exit.
Main flow: 1. The staff selects Exit Room on the touchscreen. 2. The system prompts for SID and PIN. 3. SRC verifies the SID & PIN with HRS 4. If valid, the airlock door opens. 5. the exit is recorded .
Alternative flows: 3a. Invalid SID or PIN 1. Staff is denied permission to exit. 2. The system locks the airlock until a Security Guard intervenes. 3. Use case is terminated. 3b. If Staff is the meeting chair. 1. Meeting Chair is denied exit if the meeting is still ongoing 2. Use case is terminated.

Use Case: Verify SID and PIN
ID: 3
Goal: Authenticate staff identity before allowing access.
Primary actor: Secure room controller (SRC)
Secondary actor(s): Human resource system (HRS)
Preconditions: 1. The staff must have a valid SID and PIN stored in HRS.
Postconditions: 1. If valid, the system proceeds to security level checks. 2. If invalid, access is denied.
Main flow: 1. The staff enters their SID and PIN on the touchscreen. 2. SRC sends this data to HRS for verification. 3. HRS checks the credentials and responds with either "Valid" or "Invalid."
Alternative flows: 2a. Invalid SID or PIN 1. SRC denies access. 2. SRC logs the failed attempt. 3. Use case is terminated.

Use Case: Archive Logs
ID: 4
Goal: Securely store meeting logs
Primary actor: Secure room controller (SRC)
Secondary actor(s): Meeting archive system (MAS)
Preconditions: 1. A 24-hour period must have completed. 2. Meeting logs (ML) must be available for archiving.
Postconditions: 1. Logs are successfully archived in MAS. 2. Local logs in SRC are deleted to maintain security.
Main flow: 1. At 01:59 AM, SRC prepares meeting log from the previous 24 hours. 2. SRC sends the log to MAS for storage. 3. MAS acknowledges the meeting log. 4. SRC deletes its copy of the meeting log.
Alternative flows: (None)

Use Case: Retrieve Schedules
ID: 5
Goal: Ensure SRC has the most up-to-date meeting schedule for the next 24 hours.
Primary actor: Secure room controller (SRC)
Secondary actor(s): Global meeting scheduler (GMS)
Preconditions: 1. The Global Meeting Scheduler (GMS) must be available.
Postconditions: 1. SRC has the correct schedule for the next 24 hours. 2. The system is ready to validate meeting participants.
Main flow: 1. At 02:00 AM, SRC sends a request to GMS for the next 24-hour schedule. 2. GMS responds with meeting schedules. 3. SRC stores the schedule for the meeting.
Alternative flows: (None)

Use Case: Deny Entry
ID: 6
Goal: To prevent unauthorized access to a secure room
Primary actor: Secure Room Controller (SRC)
Secondary actor(s): Human Resource system (HRS)
Preconditions: 1. The staff attempting entry has provided credentials. 2. The system has validated the credentials and security requirements.
Postconditions: 1. Entry is denied, and an attempt log is created. 2. If security concerns arise, an alert may be triggered.
Main flow: 1. SRC verifies the SID and PIN using HRS. 2. If credentials are invalid, access is denied. 3. If security clearance is insufficient, access is denied. 4. If a probationer attempts to enter a Top Secret or Secret meeting without their Line Manager, access is denied. 5. If entry is denied, it gets recorded in the ML.
Alternative flows: 5a. Silent alarm 1. Silent alarm is activated 2. Security is alerted. 3. Use case is terminated.

Use Case: When Access Mode = Staff

ID: 7

Goal: To authenticate a staff member and determine whether they meet the security requirements for entry.

Primary actor: Staff

Secondary actor(s): Human Resource system (HRS)

Preconditions:

1. The staff member must have valid credentials registered in HRS

Postconditions:

1. If verified, the staff member enters the secure room.
2. If denied, the system records the failed attempt in the log.

Main flow:

1. The staff selects Staff Access Mode on the touchscreen.
2. The system prompts for SID and PIN.
3. SRC verifies credentials via HRS .
4. If credentials are valid, the system checks security level requirements.
5. If all conditions are met, the airlock door opens.

Alternative flows:

3a. Invalid SID or PIN

1. SRC denies entry if user has invalid SID and PIN
2. Use case is terminated.

4a. Insufficient security clearance

1. SRC denies entry if required security level is not met.
2. Use case is terminated.

Use Case: Check Security Level

ID: 8

Goal: To ensure the staff member's security clearance level meets the meeting security level.

Primary actor: Secure Room Controller (SRC)

Secondary actor(s): Human Resource system (HRS)

Preconditions:

1. The staff member has entered valid credentials.
2. The system has retrieved the security clearance level from HRS.

Postconditions:

1. If access is granted, the staff member proceeds to the next step in entry.
2. If denied, the attempt is logged.

Main flow:

1. SRC retrieves the Security Clearance Level (SCL) from HRS.
2. The system checks the meeting's Security Level (SL).
3. If the SCL meets or exceeds the required SL, access is granted.

Alternative flows:

- 2a. Insufficient security clearance
1. SRC denies entry if required security level is not met.
 2. Use case is terminated.

Use Case: Verify Line Manager Presence

ID: 9

Goal: To ensure that a probationer entering a high-security meeting has their line manager present.

Primary actor: Secure Room Controller (SRC)

Secondary actor(s): Human Resource system (HRS)

Preconditions:

1. The staff member is classified as a Probationer.
2. The meeting security level is Top Secret or Secret.

Postconditions:

1. The probationer is granted access only if the Line Manager is present.
2. If access is denied, the attempt is logged.

Main flow:

1. SRC identifies that the staff member is a Probationer.
2. SRC retrieves the Line Manager SID from HRS.
3. SRC checks whether the Line Manager is present in the room.
4. If present, access is granted

Alternative flows:

3a. Probationer Handling

1. Line manager is not present, SRC denys entry.
2. Use case is terminated.

Use Case: Start Meeting
ID: 10
Goal: To officially start a meeting once the Meeting Chair (MC) enters the room.
Primary actor: Meeting Chair (MC)
Secondary actor(s): Secure Room Controller (SRC)
Preconditions: 1. Meeting is scheduled between 6:00 am to 10:00pm. 2. A scheduled meeting must exist for the current time. 3. The MC must have valid credentials and security clearance.
Postconditions: 1. The meeting is officially marked as Started. 2. The SRC records attendees.
Main flow: 1. The MC enters the room. 2. SRC verifies the MC's credentials and Security Clearance Level (SCL). 3. SRC marks the meeting as Started. 4. Entry and exit tracking begin for all participants.
Alternative flows: (None)

Use Case: When Access Mode = Security

ID: 11

Goal: To allow a Security Guard to override normal access controls and respond to security events.

Primary actor: Security Guard

Secondary actor(s): Secure Room Controller (SRC)

Preconditions:

1. A security event requires manual intervention.

Postconditions:

1. The Security Guard gains access if needed.
2. If denied, an alert is logged.

Main flow:

1. The Security Guard selects Security Mode on the external touchscreen.
2. The system prompts for a 7-character security code.
3. If valid, the security override is activated.
4. The guard gains control of the airlock doors.

Alternative flows:

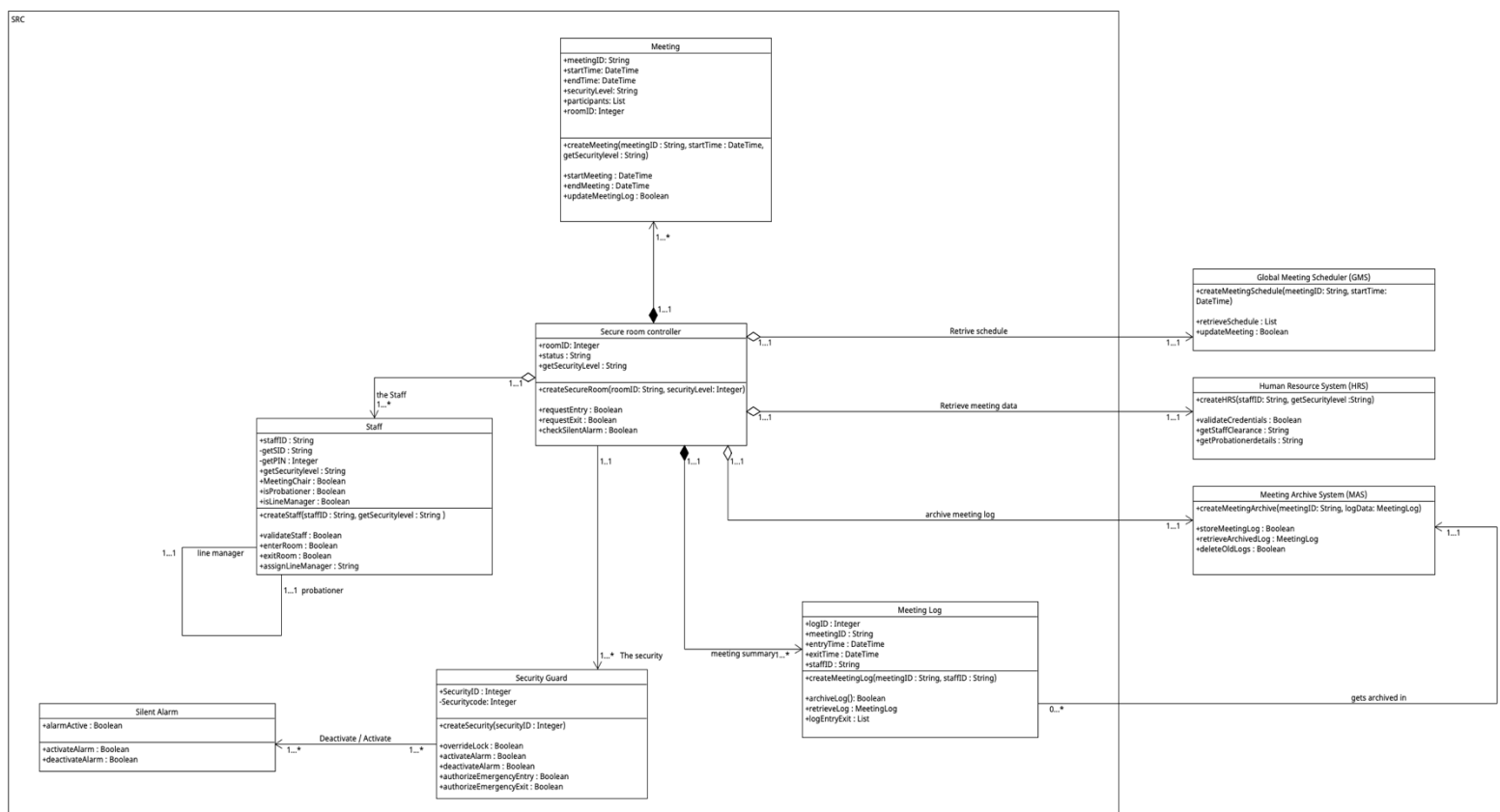
- 2a. Invalid Security Code
 1. Access is denied.
 2. Use case is terminated.

Use Case: Deactivate alarm
ID: 12
Goal: To deactivate a triggered Silent Alarm and restore normal access control.
Primary actor: Security Guard
Secondary actor(s): Secure Room Controller (SRC)
Preconditions: 1. A Silent Alarm must be active. 2. Only a Security Guard can deactivate it.
Postconditions: 1. The silent alarm is deactivated. 2. Normal access control is restored.
Main flow: 1. The Security Guard selects Deactivate Alarm on the touchscreen. 2. The system prompts for a 7-character security code. 3. SRC deactivates the silent alarm. 4. Normal access resumes.
Alternative flows: 2a. Invalid Security Code 1. Access is denied. 2. Use case is terminated.

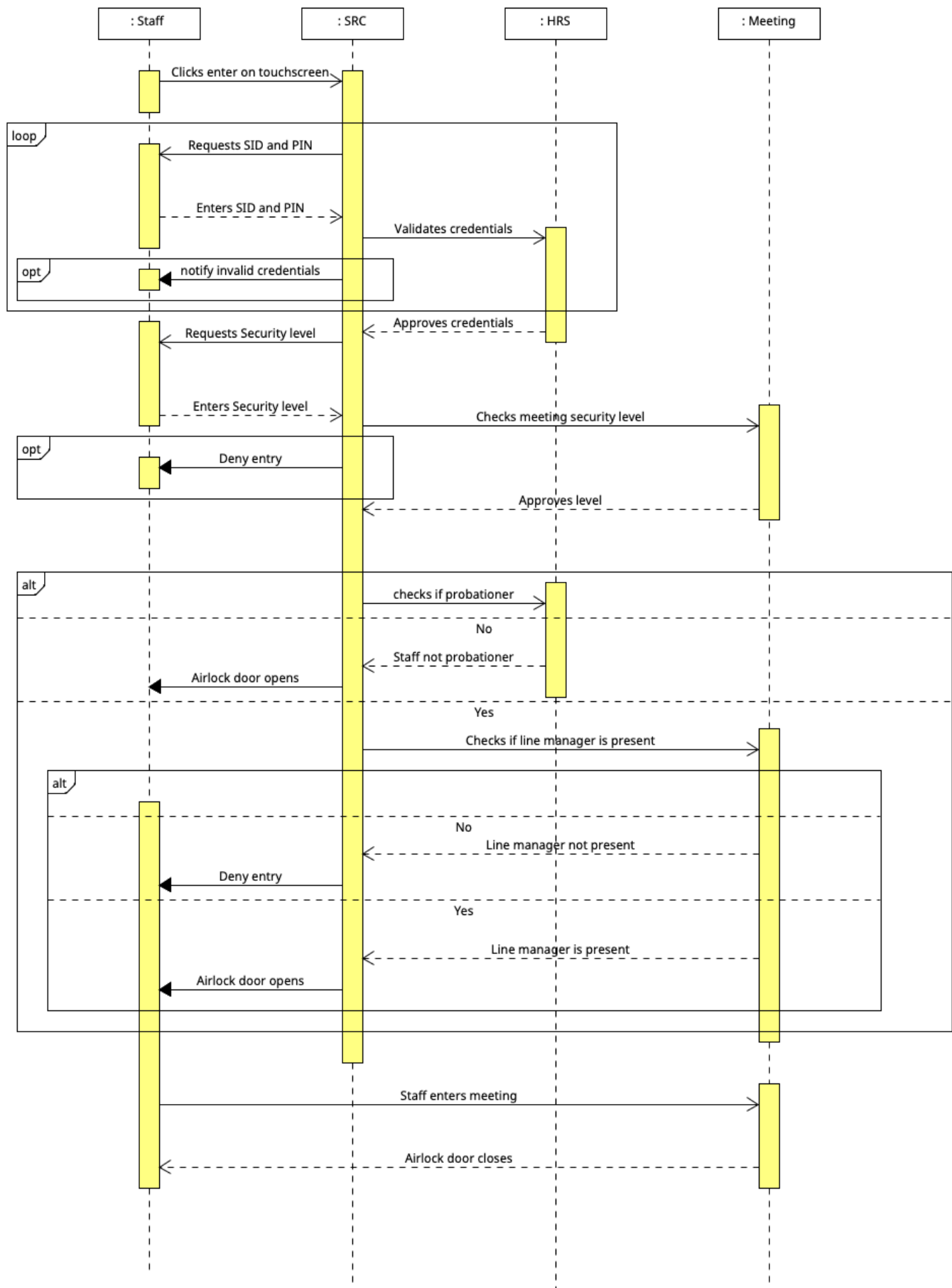
D5: Traceability Matrix

	UC1	UC2	UC3	UC4	UC5	UC6	UC7	UC8	UC9	UC10	UC11	UC12
FR1	X						X				X	
FR2	X	X	X			X	X				X	X
FR3	X					X	X	X		X		
FR4	X				X					X		
FR5	X											
FR6	X						X		X			
FR7	X						X			X		
FR8	X										X	X
FR9					X							
FR10				X								
FR11				X								
FR12										X		

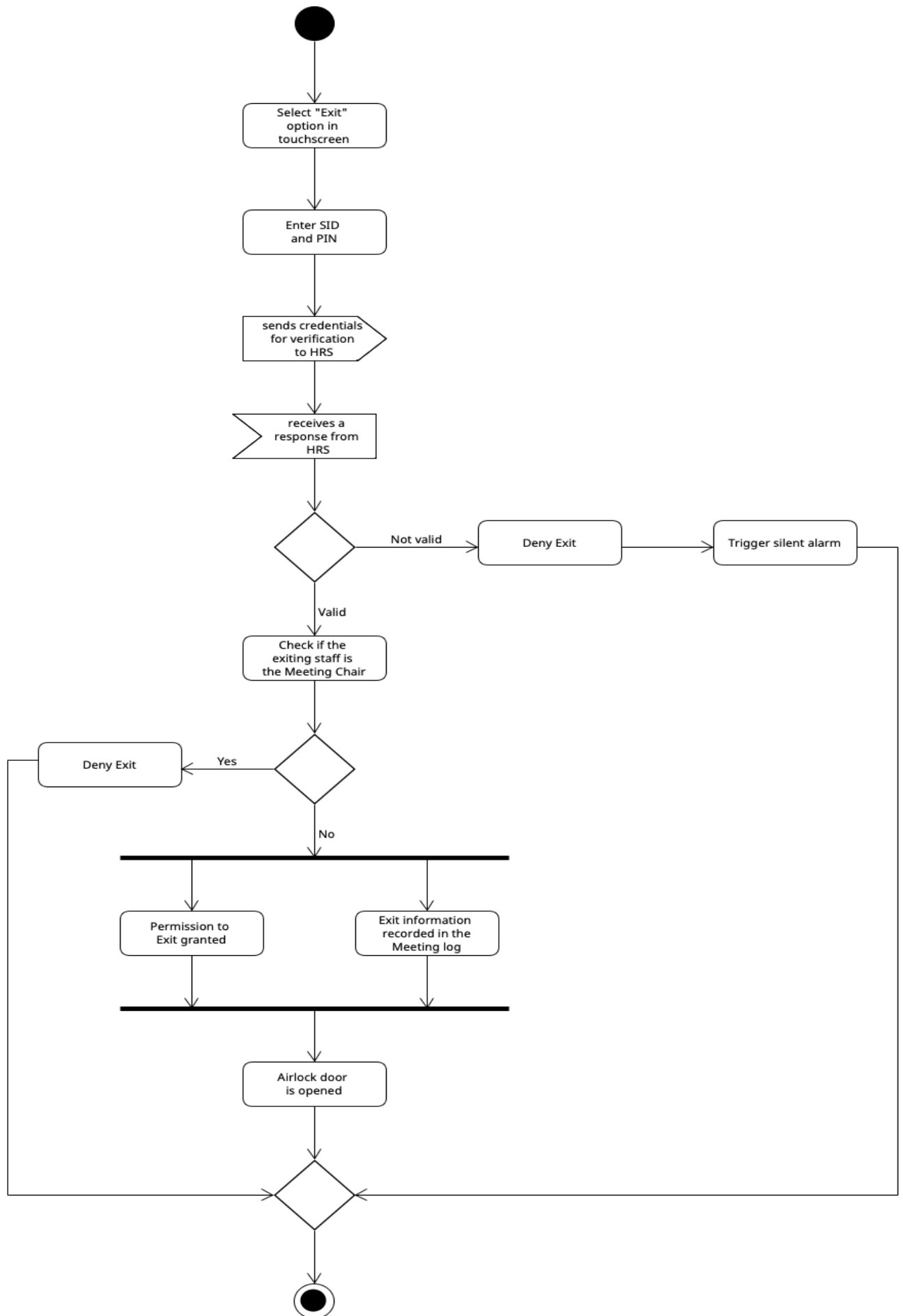
D6: Class Diagram



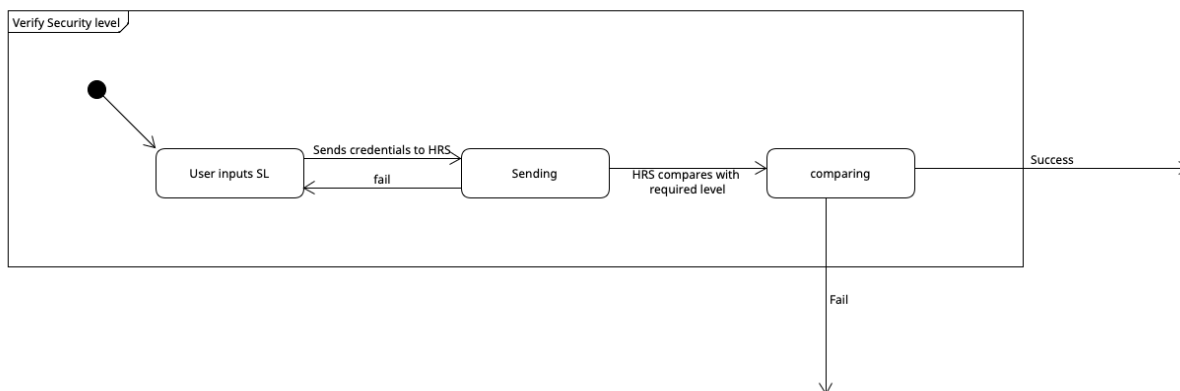
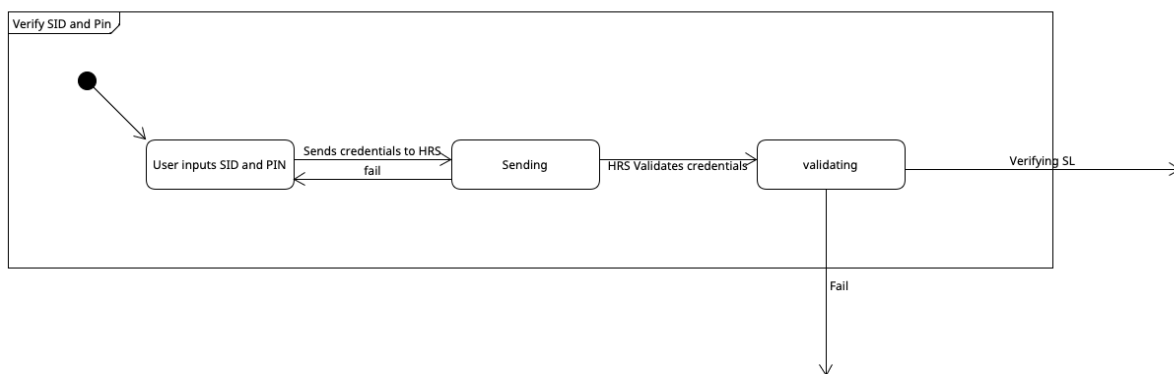
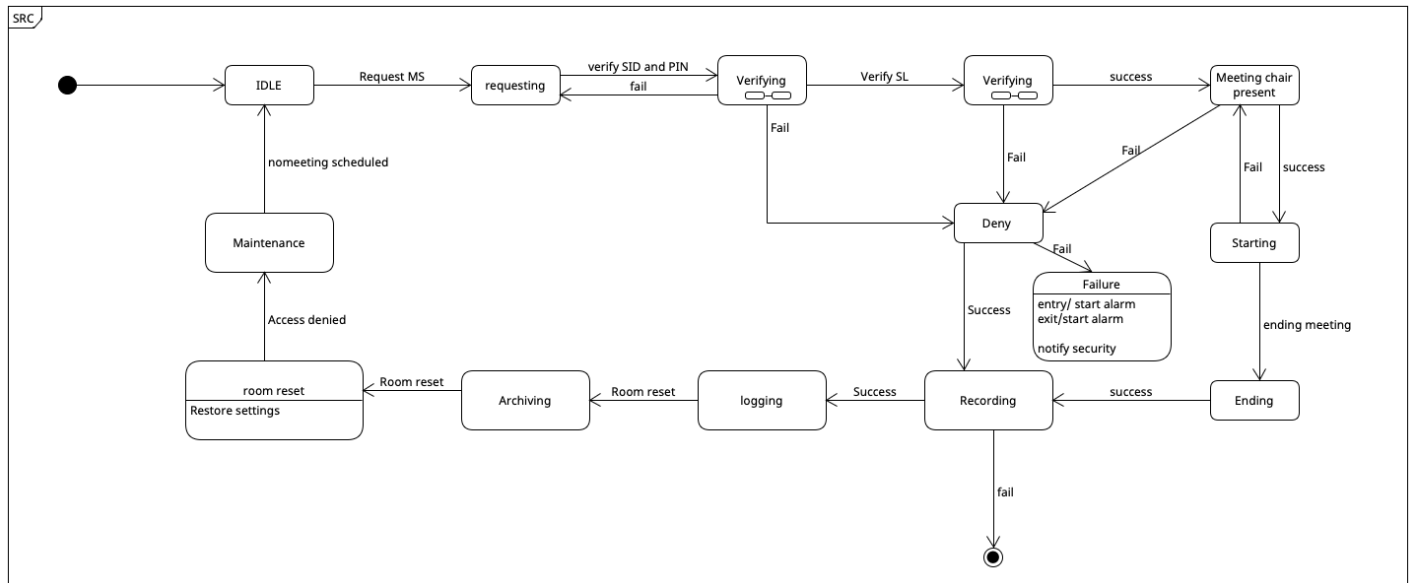
D7: Sequence diagram



D8: Activity Diagram



D9: State Machine Diagram



D10: Scenario Test cases

Input condition	Valid case	Invalid case
Staff selects "Exit" on touchscreen	"Exit" option is pressed	No action taken
SID and PIN	Valid Sid (6-character code) and Pin (4-digit code) is entered by staff	Invalid Sid and Pin is entered by staff
Response received from HRS	Valid credentials returned	Invalid credentials received
Meeting Chair Status	Exiting staff is not the Meeting Chair or meeting is over	Exiting staff is the Meeting Chair , but the meeting is still ongoing.
Exit permission granted	Permission granted	Permission denied due to failed verification
Airlock door operation	Door opens upon permission granted	Door remains locked due to failure in validation
Exit information recorded in Meeting Log	Successful log update in Meeting Archive System	Log entry failure

Test case	Test data	Expected result	Path
TC1	"Exit" option is pressed	Accept	<pre> graph TD Start(()) --> SelectExit[Select "Exit" option on touchscreen] SelectExit --> EnterSID[Enter SID and PIN] EnterSID --> SendCredentials[Send credentials for verification to HRS] SendCredentials --> ReceivedResponse{received a response from HRS} ReceivedResponse -- No --> DenyExit[Deny Exit] DenyExit --> TriggerAlert[Trigger alert alarm] ReceivedResponse -- Yes --> Valid{Valid} Valid --> CheckMeetingChair{Check if the exiting staff is the Meeting Chair} CheckMeetingChair -- No --> DenyExit CheckMeetingChair -- Yes --> ParallelSplit(()) ParallelSplit --> PermissionNotGranted[Permission is not granted] PermissionNotGranted --> DenyExit ParallelSplit --> RecordLog[Exit information recorded in the Meeting Log] PermissionNotGranted --> RecordLog RecordLog --> DoorOpened[Airlock door is opened] DoorOpened --> End(()) </pre>

TC2	Valid Sid (6-character code) and Pin (4-digit code) is entered by staff	Accept	<pre> graph TD Start(()) --> SelectTest[Select "Test" option in touch screen] SelectTest --> EnterSidPin[Enter Sid and Pin] EnterSidPin --> SendCredentials[Send credentials for verification to HRS] SendCredentials --> HRSResponse[HRS sends a response] HRSResponse --> Valid{Valid} Valid -- No --> DenyTest[Deny Test] DenyTest --> TriggerAlarm[Trigger silent alarm] Valid -- Yes --> CheckChat[Check if the testing staff is in the Meeting Chat] CheckChat -- No --> DenyTest CheckChat -- Yes --> Split(()) Split --> PushNotif[Push notification to all attendees] Split --> NextMeeting[At next meeting, testing begins] PushNotif --> Join(()) NextMeeting --> Join Join --> SendPushNotif[Send push notification to all attendees] SendPushNotif --> EndTest{ } EndTest -- No --> EnterSidPin EndTest -- Yes --> End(()) </pre>
TC3	Valid credentials is received from HRS	Accept	<pre> graph TD Start(()) --> SelectTest[Select "Test" option in touch screen] SelectTest --> EnterSidPin[Enter Sid and Pin] EnterSidPin --> SendCredentials[Send credentials for verification to HRS] SendCredentials --> HRSResponse[HRS sends a response] HRSResponse --> Valid{Valid} Valid -- No --> DenyTest[Deny Test] DenyTest --> TriggerAlarm[Trigger silent alarm] Valid -- Yes --> CheckChat[Check if the testing staff is in the Meeting Chat] CheckChat -- No --> DenyTest CheckChat -- Yes --> Split(()) Split --> PushNotif[Push notification to all attendees] Split --> NextMeeting[At next meeting, testing begins] PushNotif --> Join(()) NextMeeting --> Join Join --> SendPushNotif[Send push notification to all attendees] SendPushNotif --> EndTest{ } EndTest -- No --> EnterSidPin EndTest -- Yes --> End(()) </pre>

TC4	Exiting staff is the Meeting Chair , but the meeting is over.	Accept	<pre> graph TD Start(()) --> Select[Select "Exit" option in touchscreens] Select --> Enter[Enter ID and PIN] Enter --> Send[Send credentials for verification to HES] Send --> Receive[/receives a response from HES/] Receive --> Valid{ } Valid -- Not valid --> Deny1[Deny Exit] Deny1 --> Alarm1[Trigger silent alarm] Valid -- Valid --> CheckChair[Check if the exiting staff is the Meeting Chair] CheckChair --> Chair{ } Chair -- Yes --> Deny2[Deny Exit] Deny2 --> Alarm1 Chair -- No --> Fork(()) Fork --> Grant[Permission to Exit granted] Fork --> Record[Exit information recorded in the Meeting log] Grant --> Join(()) Record --> Join Join --> Unlock[Airlock door is opened] Unlock --> End(()) </pre>
TC5	Exiting staff is not the Meeting Chair	Accept	<pre> graph TD Start(()) --> Select[Select "Exit" option in touchscreens] Select --> Enter[Enter ID and PIN] Enter --> Send[Send credentials for verification to HES] Send --> Receive[/receives a response from HES/] Receive --> Valid{ } Valid -- Not valid --> Deny1[Deny Exit] Deny1 --> Alarm1[Trigger silent alarm] Valid -- Valid --> CheckChair[Check if the exiting staff is the Meeting Chair] CheckChair --> Chair{ } Chair -- Yes --> Deny2[Deny Exit] Deny2 --> Alarm1 Chair -- No --> Fork(()) Fork --> Grant[Permission to Exit granted] Fork --> Record[Exit information recorded in the Meeting log] Grant --> Join(()) Record --> Join Join --> Unlock[Airlock door is opened] Unlock --> End(()) </pre>

TC6	Successful log update in Meeting Log	Accept	<pre> graph TD Start(()) --> SelectExit[Select "Exit" option in touchscreen] SelectExit --> EnterSID[Enter SID and PIN] EnterSID --> SendCreds[sends credentials for verification to HRS] SendCreds --> RecvResp[/receives a response from HRS/] RecvResp --> Validity{ } Validity -- Not valid --> DenyExit1[Deny Exit] DenyExit1 --> Alarm[Trigger silent alarm] Validity -- Valid --> CheckChair[Checks if the exiting staff is the Meeting Chair] CheckChair --> ChairStatus{ } ChairStatus -- Yes --> DenyExit2[Deny Exit] ChairStatus -- No --> Split1[] Split1 --> PermExit[Permission to Exit granted] Split1 --> LogExit[Exit information recorded in the Meeting log] PermExit --> Join1[] LogExit --> Join1 Join1 --> DoorOpen[Airlock door is opened] DoorOpen --> FinalDecision{ } FinalDecision -- Yes --> DenyExit3[Deny Exit] DenyExit3 --> Alarm FinalDecision -- No --> End(()) </pre>
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Test case	Test data	Expected result	Path
TC7	4-character code SID and 4 digit code PIN is enter by staff	Reject	<pre> graph TD Start(()) --> SelectExit[Select "Exit" option in touchscreen] SelectExit --> EnterSID[Enter SID and PIN] EnterSID --> SendCreds[sends credentials for verification to HRS] SendCreds --> RecvResp[/receives a response from HRS/] RecvResp --> Validity{ } Validity -- Not valid --> DenyExit1[Deny Exit] DenyExit1 --> Alarm[Trigger silent alarm] Validity -- Valid --> CheckChair[Checks if the exiting staff is the Meeting Chair] CheckChair --> ChairStatus{ } ChairStatus -- Yes --> DenyExit2[Deny Exit] ChairStatus -- No --> Split1[] Split1 --> PermExit[Permission to Exit granted] Split1 --> LogExit[Exit information recorded in the Meeting log] PermExit --> Join1[] LogExit --> Join1 Join1 --> DoorOpen[Airlock door is opened] DoorOpen --> FinalDecision{ } FinalDecision -- Yes --> DenyExit3[Deny Exit] DenyExit3 --> Alarm FinalDecision -- No --> End(()) </pre>

TC8	6-character code SID and 2 digit code PIN is enter by staff	Reject	<pre> graph TD Start(()) --> Select[Select "Exit" option in touchscreen] Select --> Enter[Enter SID and PIN] Enter --> Send[send credentials for verification to HRS] Send --> Receive[/receives a response from HRS/] Receive --> Validity{ } Validity -- Not valid --> Deny1[Deny Exit] Deny1 --> Alarm[Trigger silent alarm] Validity -- Valid --> Check[Check if the exiting staff is the Meeting Chair] Check --> Meeting{ } Meeting -- Yes --> Deny2[Deny Exit] Deny2 --> Alarm Meeting -- No --> Split1[] Split1 --> Permit[Permission to Exit granted] Split1 --> Log[Exit information recorded in the Meeting log] Permit --> Split2[] Log --> Split2 Split2 --> Open[Airlock door is opened] Open --> Exit{ } Alarm --> Exit Deny2 --> Exit Exit --> End(()) </pre>
TC9	Invalid Sid (6-character code) and Pin (4-digit code) is entered by staff	Reject	<pre> graph TD Start(()) --> Select[Select "Exit" option in touchscreen] Select --> Enter[Enter SID and PIN] Enter --> Send[send credentials for verification to HRS] Send --> Receive[/receives a response from HRS/] Receive --> Validity{ } Validity -- Not valid --> Deny1[Deny Exit] Deny1 --> Alarm[Trigger silent alarm] Validity -- Valid --> Check[Check if the exiting staff is the Meeting Chair] Check --> Meeting{ } Meeting -- Yes --> Deny2[Deny Exit] Deny2 --> Alarm Meeting -- No --> Split1[] Split1 --> Permit[Permission to Exit granted] Split1 --> Log[Exit information recorded in the Meeting log] Permit --> Split2[] Log --> Split2 Split2 --> Open[Airlock door is opened] Open --> Exit{ } Alarm --> Exit Deny2 --> Exit Exit --> End(()) </pre>

TC10	Invalid credentials is received from HRS	Reject	<pre> graph TD Start(()) --> Select[Select "Exit" option in touchscreen] Select --> Enter[Enter ID and PIN] Enter --> Send[send credentials for verification to HRS] Send --> Receive[/receive a response from HRS/] Receive --> Valid{Valid} Valid -- Not valid --> DenyExit1[Deny Exit] DenyExit1 --> Alarm1[Trigger silent alarm] Valid -- Valid --> CheckChair[Check if the exiting staff is the Meeting Chair] CheckChair --> Chair{ } Chair -- Yes --> DenyExit2[Deny Exit] Chair -- No --> Split1[] Split1 --> Perm[Permission to Exit granted] Split1 --> Log[Exit information recorded in the Meeting log] Perm --> Merge1[] Log --> Merge1 Merge1 --> OpenDoor[Airlock door is opened] OpenDoor --> DoorOpen{ } DoorOpen -- No --> DenyExit3[Deny Exit] DenyExit3 --> Alarm2[Trigger silent alarm] DoorOpen -- Yes --> End(()) </pre>
TC11	Exiting staff is the Meeting Chair, but the meeting is still ongoing.	Reject	<pre> graph TD Start(()) --> Select[Select "Exit" option in touchscreen] Select --> Enter[Enter ID and PIN] Enter --> Send[send credentials for verification to HRS] Send --> Receive[/receive a response from HRS/] Receive --> Valid{Valid} Valid -- Not valid --> DenyExit1[Deny Exit] DenyExit1 --> Alarm1[Trigger silent alarm] Valid -- Valid --> CheckChair[Check if the exiting staff is the Meeting Chair] CheckChair --> Chair{ } Chair -- Yes --> DenyExit2[Deny Exit] Chair -- No --> Split1[] Split1 --> Perm[Permission to Exit granted] Split1 --> Log[Exit information recorded in the Meeting log] Perm --> Merge1[] Log --> Merge1 Merge1 --> OpenDoor[Airlock door is opened] OpenDoor --> DoorOpen{ } DoorOpen -- No --> DenyExit3[Deny Exit] DenyExit3 --> Alarm2[Trigger silent alarm] DoorOpen -- Yes --> End(()) </pre>